

PROSTHETIC SENSOR SOCK

A PROJECT90 CREATION

TECH FUTURES CHALLENGE 2026

Prosthetic Design is a Long Process

Relying on qualitative data alone means months of appointments before a comfortable fit



EXISTING TOOLS DO NOT CAPTURE THE FULL STORY



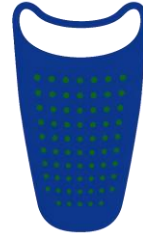
PATIENT FEEDBACK

is subjective and invites
miscommunication due to pain
and frustration.



PHANTOM PAIN

can make patient-reported
discomfort harder to interpret
during fitting



PREVIOUS SOLUTIONS

may be costly, wired, or difficult to
integrate into routine fitting

3 PRIMARY OBJECTIVES



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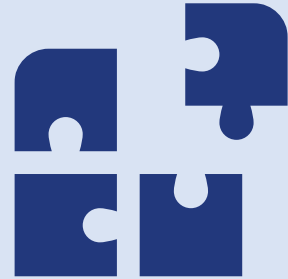
CONSISTENT READINGS

Capture repeatable
signals and display
them clearly



COST EFFECTIVE

Improved value for cost
for full adoption



SIMPLE INTEGRATION

Easy for clinicians and
patients to understand

OUR SOLUTION: COST-EFFECTIVE AND WIRELESS



FSR-400 SENSORS

30-GAUGE WIRE

MULTIPLEXER

ESP32 / PCB

MATLAB HEATMAP

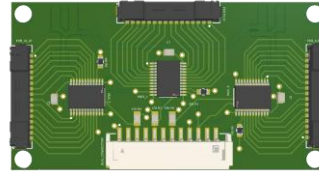
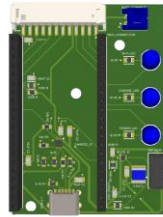
Physical sock + electronics + wireless visualization

OUR DATA PIPELINE FROM LIMB TO COMPUTER



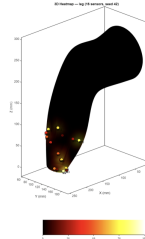
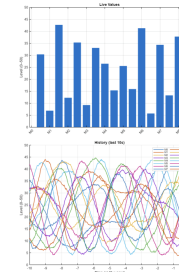
Force Sensitive Resistor

Applying pressure
increases voltage



PCB

Analog to Digital
conversion

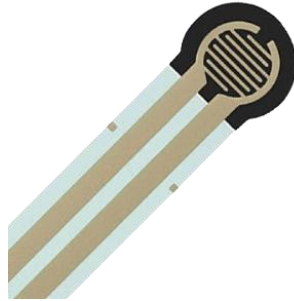


MATLAB

Live 3D heatmap
visualization

REAL-TIME PROTOTYPE PRESSURE VISUALIZATION.

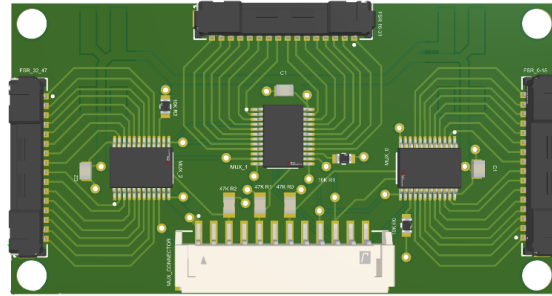
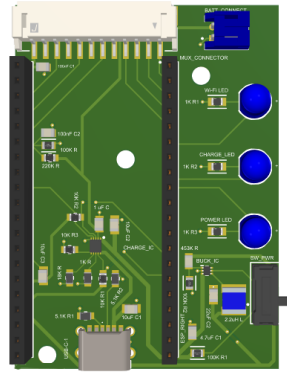
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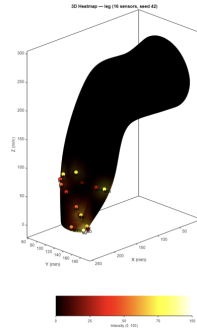
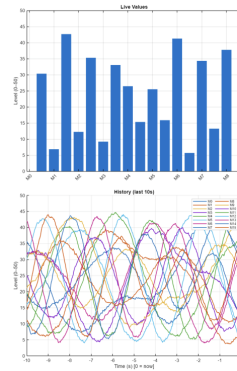


PCB

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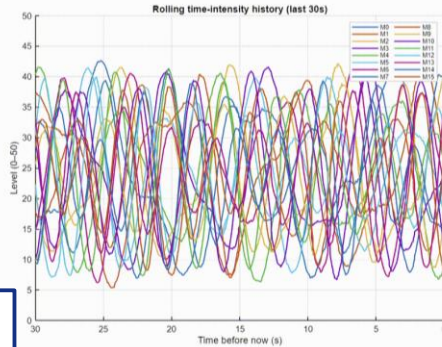
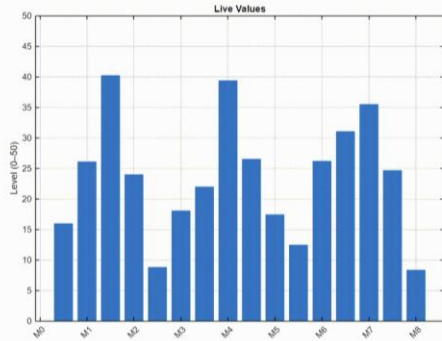
MATLAB

**Live 3D heatmap
visualization**

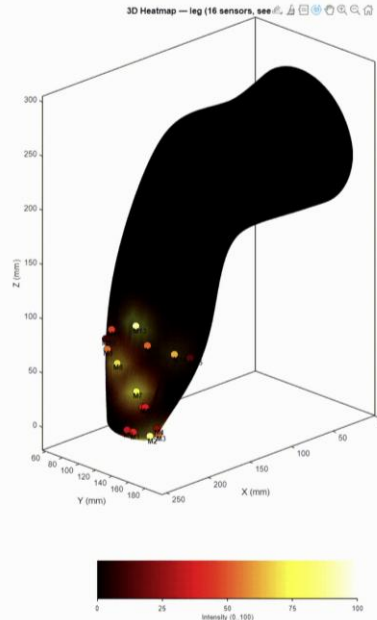
REAL-TIME PROTOTYPE PRESSURE VISUALIZATION.

DATA VISUALIZATION THAT WORKS

**Real-Time
Values**

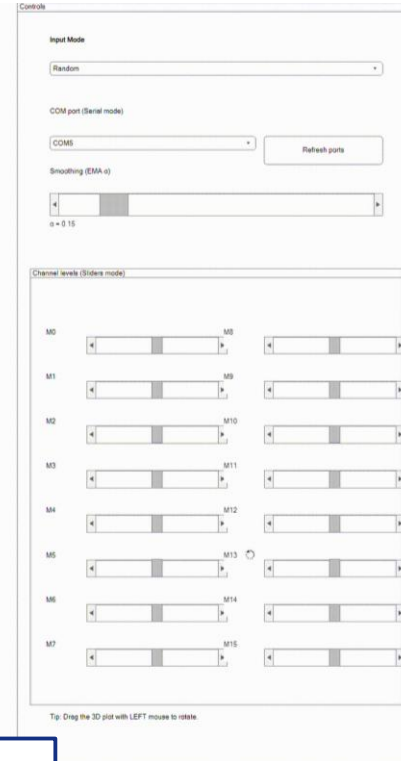


**Historical
Values**



3D Heatmap

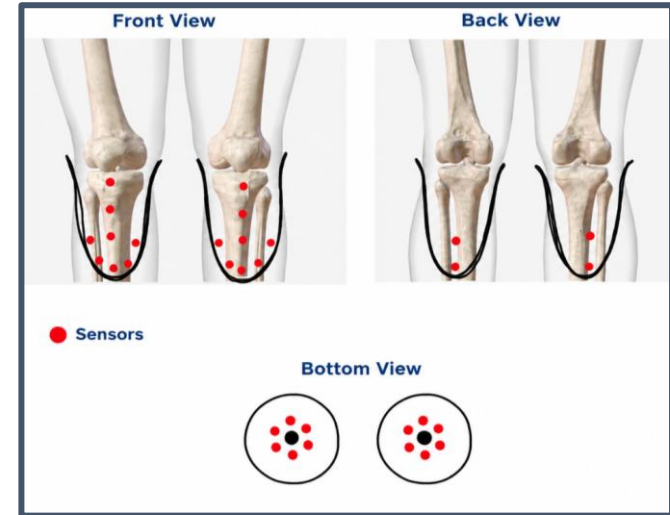
**Program
Controls**



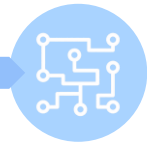
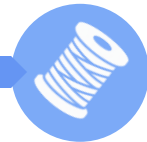
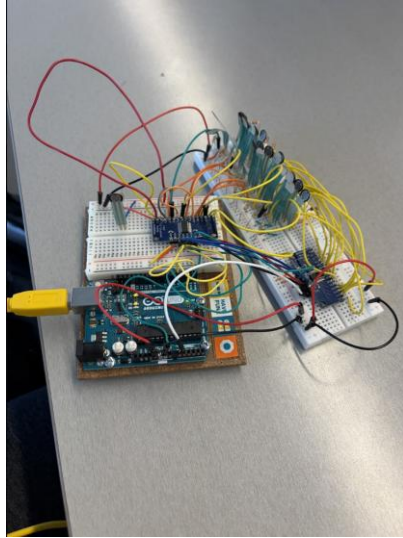
PHYSIOLOGICAL RELEVANCE

■ Why These Locations?

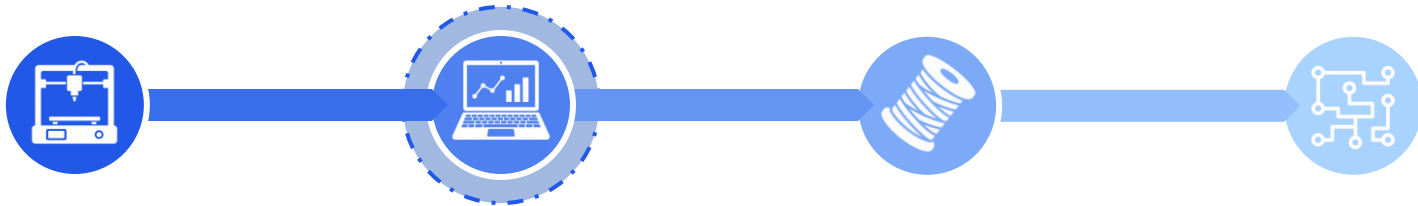
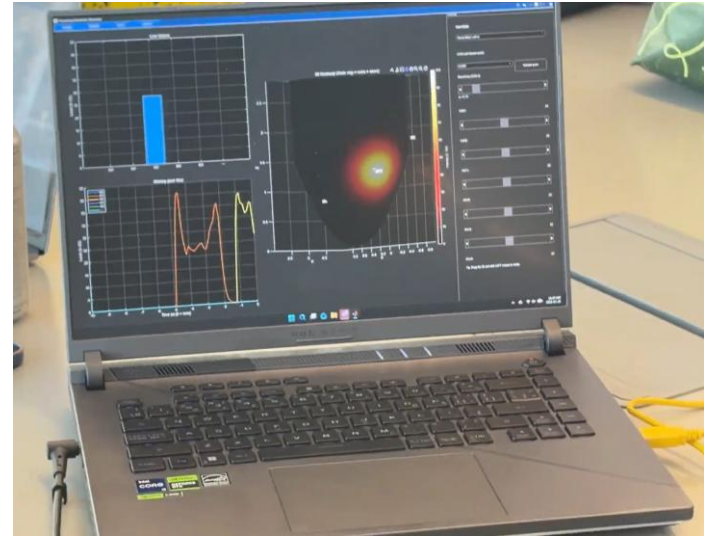
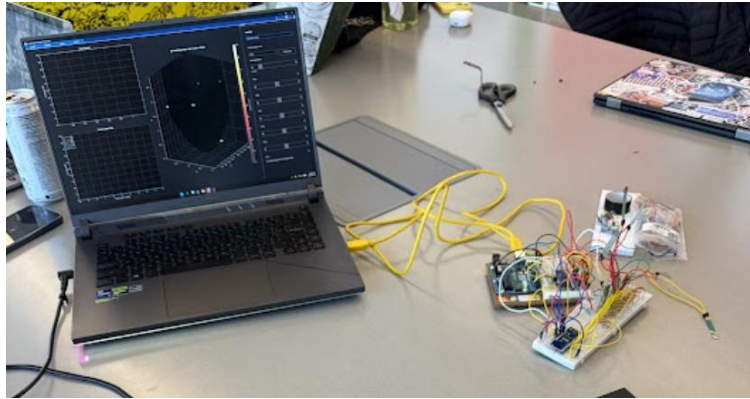
- Transtibial (below-knee) focus -> tibia + fibula are primary load-bearing bones
- Key pressure-sensitive landmarks:
 - Tibial crest
 - Distal tibia
 - Fibular head
 - Distal fibula
 - Residual limb end
- Low soft tissue coverage -> higher interface pressure risk



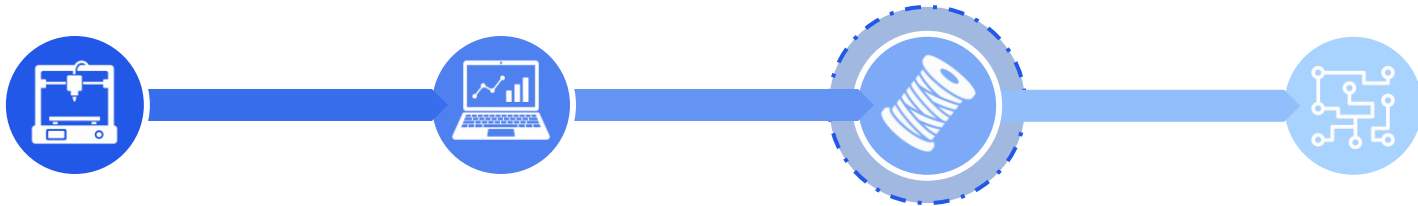
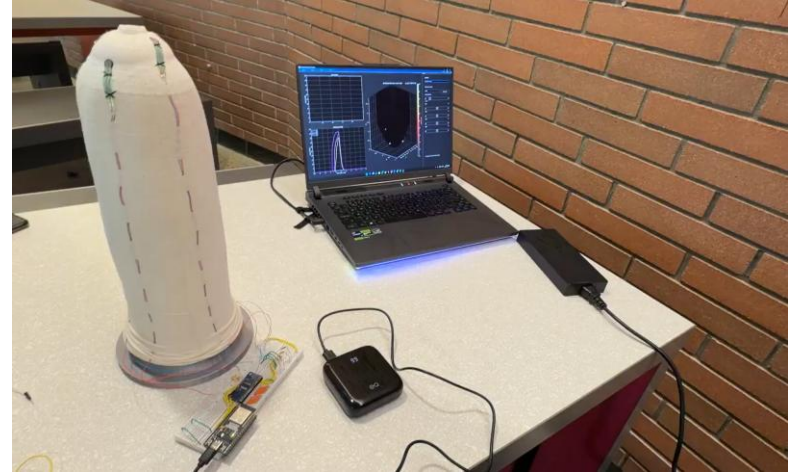
DESIGN ITERATIONS



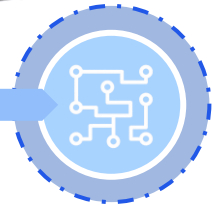
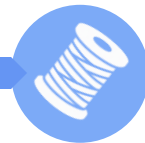
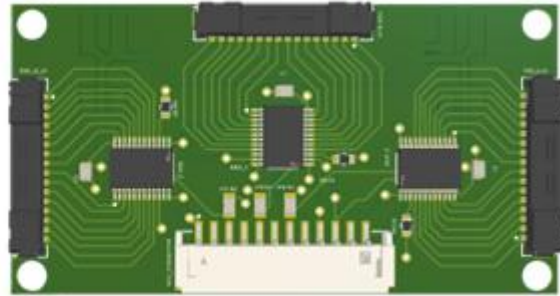
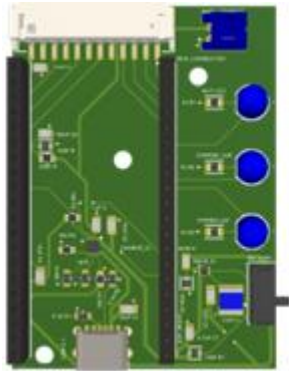
DESIGN ITERATIONS



DESIGN ITERATIONS



DESIGN ITERATIONS



THREE THINGS WE LEARNED

01

FIRST PRINCIPLES ENGINEERING

- Rudimentary solutions proved effective
- Niche methods were difficult to integrate

02

HEALTHCARE FOCUSED DESIGN

- Keeping patient experience in mind
- Comfort matters as well as safety
- Easy interpretation for clinical use

03

SEEKING MENTOR FEEDBACK

- Discussions with clinics and labs grounded our project scope
- Set practical standards for the prototype design

FUTURE GOALS

■ PCB INTEGRATION

- Transition from breadboard to custom PCB for stability
- Ensure long term reliability of sensors

■ Load Testing

- Pursue ethics-approved testing with prosthetic users
- Creating a standard curve for load values

■ Use in Clinics

- Evaluate the software in clinical research settings to support fitting decisions.
- Incorporate save and replay functions for deeper analysis

SOCIAL IMPACT

Aligning with SDG 3: Good Health and Well-being by exploring accessible pressure-sensing research tools for people with limb loss.

THANK YOU.

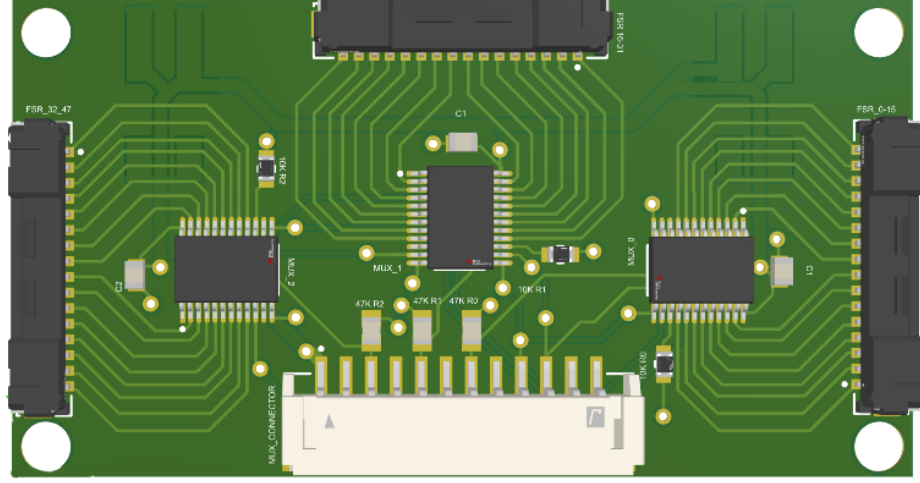
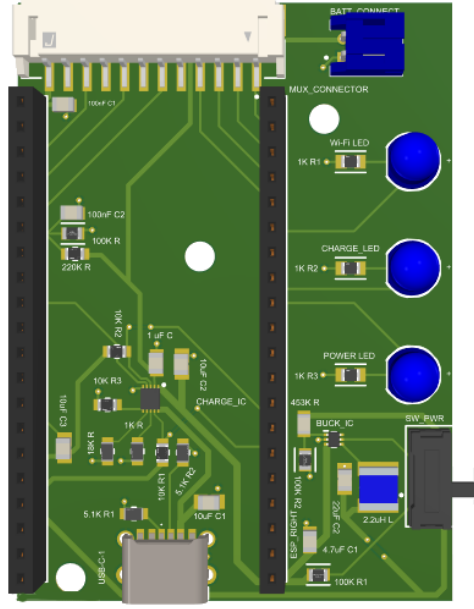
LET'S REDEFINE THE FIT

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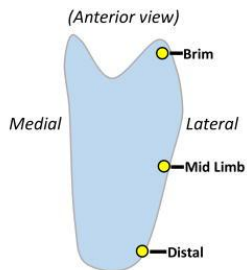
Neeor, Karlyn, Hira, Abdul



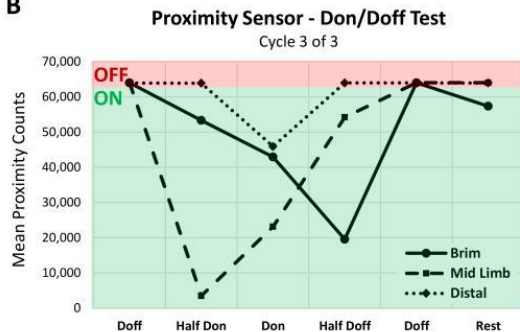
Appendix – Q&A



A



B



C

	Brim			Mid Limb			Distal		
	Cycle 1	Cycle 2	Cycle 3	Cycle 1	Cycle 2	Cycle 3	Cycle 1	Cycle 2	Cycle 3
Doff 1	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF
Half Don	ON	ON	ON	ON	ON	ON	OFF	ON	OFF
Don	ON	ON	ON	ON	ON	ON	ON	ON	ON
Half Doff	ON	ON	ON	ON	ON	ON	OFF	OFF	OFF
Doff 2	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF
Rest on Brim	ON	ON	ON	OFF	OFF	OFF	OFF	OFF	OFF

Literature Example

Industry Example



Appendix – Preliminary cost comparison

Solution	Type	Cost information	Real-time?	Requires Specialist?
Tekscan F-Socket	Clinical research system	Vendor quote required	Yes	Yes
Rincoe Socket Fitting System	Clinical system	Vendor quote required	Yes	Yes
3D scan + manual assessment	Geometry only, no pressure	Varies by provider	No	Yes
Patient verbal feedback	Subjective	No equipment cost	No	No
Our Sensor Sock	Wearable FSR prototype	Prototype estimate	Yes	No